

- 3 (a) State the principle of conservation of momentum.

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.....

..... [2]

- (b) An object of mass $2m$ is travelling at a speed of 5.0 m s^{-1} in a straight line. It collides with an object of mass $3m$ which is initially stationary, as shown in Fig. 3.1.

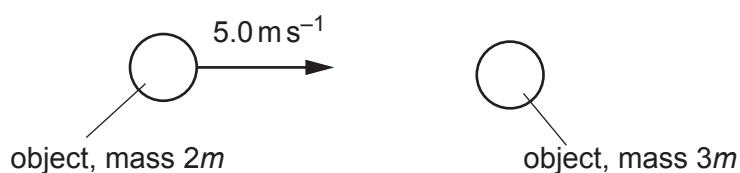


Fig. 3.1

After the collision, the object of mass $2m$ moves with velocity v at an angle of 30° to its original direction of motion.

The object of mass $3m$ moves with velocity w also at an angle of 30° , as shown in Fig. 3.2.

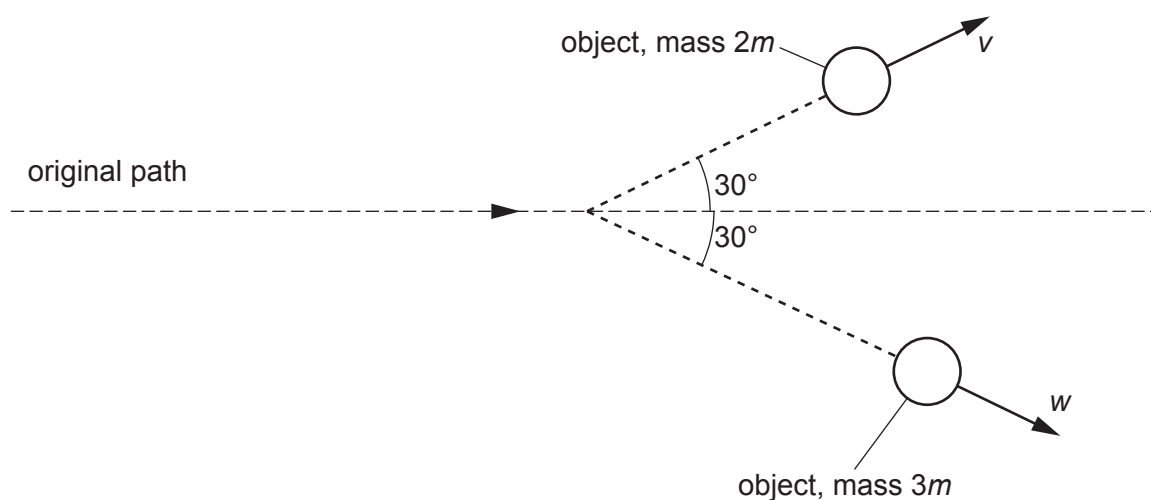


Fig. 3.2

By considering the conservation of momentum in two dimensions, calculate the magnitudes of v and w .

$$v = \dots\dots\dots \text{ms}^{-1}$$

$$w = \dots\dots\dots \text{ms}^{-1}$$

[4]

- (c) An object of mass 4.2 kg is travelling in a straight line at a speed of 6.0 ms^{-1} . The object is brought to rest in a distance of 0.050 m by a constant force.

Calculate the magnitude of this force.

$$\text{force} = \dots\dots\dots \text{N} \quad [3]$$

[Total: 9]